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A lamellar-morphology-based computational modeling for predicting the thermal conductivity of semicrystalline polymers

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ABSTRACT

Molecular-scale design of crystal structures is an emerging approach for significantly improving the intrinsic properties of polymers. In this study, we performed computational modeling to quantitatively predict the thermal conductivity of polymers based on their crystal morphology. Polyethylene lamellae with alternating crystalline and amorphous phases were prepared using isothermal crystallization within a coarse-grained molecular dynamics framework. The crystalline bulk, transient, and amorphous regions were clearly distinguished based on the distribution of the Steinhardt-bond order parameter. The thermal conductivity of the local regions was estimated discretely by applying a steady heat flow, facilitating a bottom-up prediction of the thermal response depending on the morphology of the lamellae. In particular, the integration with homogenization theory can evaluate the effective thermal conductivity of the system in terms of crystallinity, temperature, lamella thickness and orientation. The modeling successfully provided quantitative predictions for two representative hierarchical structures: spherulites and oriented crystals. Therefore, this study serves as a theoretical guide for molecular-level rational design of semicrystalline materials with high thermal conductivity.

1. Introduction

Polymers are ubiquitous in modern society because of their outstanding properties such as lightness, durability, flexibility, corrosion resistance, and facile processability [1,2]. These distinctive features arise from the conformational uniqueness of polymers, which emanates from the presence of both crystalline and amorphous phases. In particular, precisely tuning the microphase of polymers during synthesis can further diversify their functionalities and properties, enabling a wide range of engineering applications.

Conventional polymers typically exhibit a low thermal conductivity, approaching that of insulating materials [3,4]. Therefore, to improve their thermal properties, the development of polymer composite materials by incorporating dissimilar materials is considered a vital strategy [5–15]. However, if the polymer conformation is accurately controlled down to the molecular scale, for example by fabricating a single-crystal structure, the resulting phonon transport properties can far exceed the inherent limitations of prevailing amorphous or semicrystalline polymers [16–23]. This suggests that the thermal conductivity of polymers can be tuned to the desired level within a fairly large range by achieving a well-designed polymer morphology. Such innovative concepts have enabled the expansion of many engineering fields by adding functionality similar to metals or ceramics to the inherent advantages of pure polymers. Representative applications include packaging for light-emitting diodes (LEDs), corrosion-resistant heat exchangers, thermal management systems for soft electronics, and aerospace equipment enclosures [24–33].

Numerous efforts have been devoted to identifying the physicochemical factors that affect the thermal conductivity of semicrystalline polymers. The phonon transport through the backbone chains is the key energy carrier in the typical polymers [34]. In this regard, thermal conductivity along a reference direction is related to the statistically averaged spatial orientation of the molecular chain. Therefore, localizing the heat-flow direction by guiding the polymer chains through mechanical stretching, elongation, or interaction with heterogeneous interfaces is a representative strategy for modulating thermal conductivity [35–37]. Additionally, the rigidity of the polymer backbone has a significant impact on thermal conductivity. Modulation of thermal conductivity with respect to backbone stiffness comes from two main reasons [38–43]. (1) At higher backbone stiffness, phonons scattering is

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significantly reduced as the molecular freedom (including bending and torsional freedom) and segmental rotations are constrained. (2) A stiff extends the persistent length of polymers, providing an effective thermal transport path. Such influence of molecular straightness on thermal properties was also highlighted in other studies of trans-to-gauche structural transitions [44,45]. Furthermore, several studies highlighted the importance of noncovalent interactions for increasing the thermal conductivity [46,47]. Feng et al. clarified the positive correlation between the molar fraction of charged atoms and thermal properties in amorphous polymer [48]. Similarly, Yang et al. demonstrated the role of intermolecular π–π stacking in improving the axial thermal transport by suppressing the free rotation of the backbone chains and forming a long-range arrangement [49].

In semicrystalline systems, more complex factors determine the thermal properties of the polymer. As semicrystalline polymers are composed of multiple phases, including amorphous, crystalline, and intermediate regions, their microscopic composition significantly affects the resultant thermal properties [50]. In this regard, Choy et al. examined the thermal conductivity of polymer fibers drawn to various ratios, noting that the thermal properties varied with the crystallinity of the microstructure [51–53]. Therefore, strategies to maximize crystalline phase content, including electrospinning, are promising for the design of materials with metal-like properties [54]. Moreover, folded or bent chains at the ends of the polymer lamellae cause strong local phonon scattering. Duan et al. demonstrated that even a single kink in a molecule can scatter more than half of the phonons penetrating a polymer chain [55]. Therefore, expanding the effective thickness of the lamellae by controlling the pressure and cooling rate during crystallization is an intuitive strategy to mitigate the surface effect. Furthermore, local bottlenecks in heat transport can be address through the topological design of interlamellar amorphous chains, which are divided into bridges, loops, and tails. He and Liu noted that as the density of bridged chains increased, the thermal resistance of the intermediate region decreased, as further accentuated by the applied stretching strain [56]. Such importance of thermal resistance in the immediate region was also emphasized in other experimental literatures [57–59].

Understanding the structural complexity of polymers across multiple scales, including atomic, nano, and micro domains, is critical in engineering their heat-transfer properties. Depending on their conformation, polymer chains form diverse hierarchical structures from amorphous to tightly folded crystalline configurations and spherulite aggregates. Therefore, the apparent physical properties of a polymer system, including the thermal conductivity, can be dictated by the orientation and configuration of these microstructures. Despite the aforementioned efforts, a quantitative connection between the molecular-scale crystal structures and themacroscopic heat-transfer properties is still unavailable, because molecular-scale correlations between the crystalline morphological characteristics and anisotropic thermal conductivity have not yet been established. Specifically, the conformation of the polymer chain provides directionality to the heat flow and determines the macroscopic thermal conductivity of the entire system has not been clarified. Nevertheless, theoretical methodologies that can capture the molecular-level morphological features of crystallites can be utilized to identify the physicochemical factors that are critical in substantially improving heat flow in polymers.

In the present study, a theoretical modeling was devised for predicting the thermal conductivity of polymers based on their crystalline morphologies. Using coarse-grained molecular dynamics (MD) simulations, a polyethylene (PE) lamellar microstructure with alternating crystalline and amorphous phases was constructed in the nanoscale domain. The bulk and transient regions of the crystalline phase were clearly distinguished using the Steinhardt bond order parameter, thereby helping evaluate the local thermal conductivity responses to an applied constant heat flow. The directional thermal conductivity and anisotropy of the polymer lamellae were estimated based on the exposure temperature, crystal thickness, and orientation angle. Consequently, the thermal properties of two hierarchical structures—spherulites and oriented crystals—were evaluated in terms of crystal morphology using the homogenization theory. Overall, the established thermal conductivity modeling was found to yield quantitative predictions with high accuracy for a range of crystal structure factors.

The paper is outlined as follows. In Section 2, the preparation of a representative PE lamellar model with alternating amorphous and crystalline phases was highlighted. The order parameters to quantify spatial chain alignment and methodology for the thermal-conductivity measurements were also described. In Section 3, the structural and thermal evaluation of the implemented lamellar model was discussed. Parallel and perpendicular thermal conductivities depending on the order parameter were derived and extended to account for the three phases of polymer crystals, i.e., transient–bulk–transient. The quantitative predictions of the thermal conductivity of two crystal growth morphologies, spherulite and oriented crystals, were also explained. Finally, the conclusions of this study are presented in Section 4.

2. Models and methods

This section described the methodology for constructing a PE lamellar model with coexisting crystalline and amorphous regions. The detailed computational parameters, including interatomic potential and simulation software, were provided. The crystalline and amorphous regions within the nanodomain were rigorously characterized using structural-order parameters. The protocol for evaluating the global and local thermal conductivity under an applied steady-state heat was also provided.

2.1. Construction of coarse-grained polymer lamellae

A linear PE chain comprising 1000 repeat units (C_2H_4) was prepared in an MD environment. The molecules used in this study were treated as coarse-grained atoms, with each C_2H_4 unit regarded as a single particle with a molar mass of 28.05 g/mol (Fig. 1(a)). Initially, 14 molecules were placed randomly in a periodic unit cell. To avoid unintended intermolecular interference during positioning, the unit cell was set sufficiently large (density: 0.1 g/cc) (Fig. 1(b)). The internal potential energy of the entire system was minimized using the conjugated gradient method with an energy cutoff of 10 kcal/mol. The unit cell was then progressively condensed using the isothermal–isobaric (NPT) ensemble at 300 K and 1 atm for 10 ns. The motion of the constituent atoms was monitored using the Verlet algorithm with a time step of 1 fs.

A periodic lamellar structure was constructed by simulating isothermal crystallization (Fig. 1(c)). To that end, the equilibrated unit cell was first exposed to a high temperature (700 K) for 5 ns in the NPT ensemble (the conformation changes are presented in Fig. S1). Because this temperature far exceeded the typical melting point of PE (400–410 K) [60,61], the polymer chains in the unit cell became completely amorphous. The model was then stabilized for 100 ns at a constant temperature of 390 K. As the intermolecular interaction energy surpassed the thermodynamic energy, a regularly arranged microstructure was generated in the unit cell. The conformation of the resulting PE chains is further explored in Section 3.1.

The molecular behavior of the coarse-grained chains was described using a set of potential tables proposed by Moyassari et al. [62,63]. The bond, angle, and noncovalent pair potentials of the polymer chain obtained in the all-atomistic system are derived as coarse-grained potentials using the Iterative Boltzmann Inversion method. In general, the interatomic forces in polymers depend on given thermodynamic environments. Therefore, specialized potential tables describing for temperatures at 300, 350, and 450 K were individually proposed to accurately describe the molecular behavior under various thermodynamic conditions. Each potential is suitable for ±50 K of the temperature of the reference atomistic system. In this study, isothermal crystallization at 390 K was described using potential tables determined

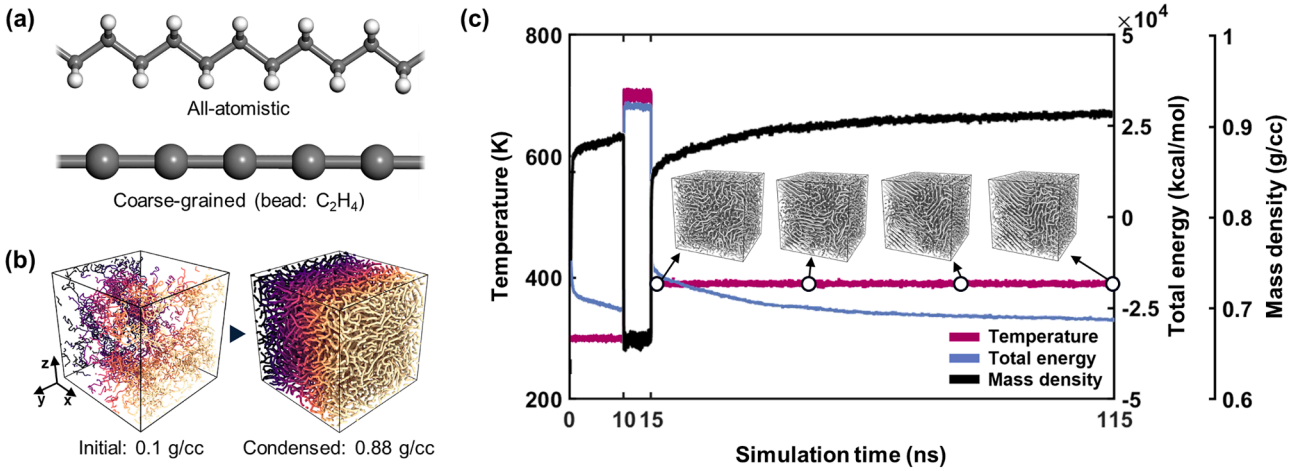


Fig. 1. Coarse-grained modeling of polymer lamellar microstructure. (a) Coarse-grained mapping of a linear PE chain. (b) Changes in the molecular configuration during the condensation. (c) Temperature, total potential energy, and density profiles during annealing and isothermal crystallization.

at 350 K, and thermal conductivity measurements were performed using one determined at 300 K. This potential corresponds well with the experimental observations for the polymer crystal morphology and mechanical properties [62,63]. Moreover, we separately reproduced the thermal properties of the polymer system obtained using the corresponding potentials and verified that the values agree well with the experimental measurements [64,65] (Supplementary Note 1 and Fig. S2).

The commercial MD software Materials Studio 2020 (Dassault Systèmes, France) was used to model the unit cell with a lamellar structure. The open-source MD code LAMMPS (Sandia Labs, USA) was used to monitor the progress in crystallization of the lamellar structure and perform the subsequent thermal conductivity measurements.

2.2. Evaluation of local order in modeled polymer crystals

The degree of polymer backbone ordering and the evolution of the crystalline phase were quantified using the Steinhardt bond order parameter ($q_6q_6^*$) as proposed by Zhang and Larson [66]. The $q_6q_6^*$ parameter is a correlation function that calculates the value for a reference atom based on the local arrangement of its adjacent atoms. The parameter is close to zero for the liquid-like amorphous phase and positive for crystalline atoms. The process for quantifying this value is described henceforth: First, the q_{6m} complex vector for the ith atom constituting the backbone chain is expressed as [67]

$$q_{6m}(i) = \frac{1}{N_i}\sum_{j=1}^{Ni} Y_6^m\left(\phi_{i,j}, \varphi_{i,j}\right), \tag{1}$$

where N_i is the number of neighboring atoms within a cutoff distance of 0.54 nm from the atom i, Y_6^m is the mth component of the sixth-degree spherical harmonic, and ($\phi_{i,j}$, $\varphi_{i,j}$) represents the spherical polar orientation of a vector connecting the atoms i and j. Subsequently, the bond order parameter $q_6q_6^*$ of atom i is calculated as follows:

$$q_6q_6^*(i) = \sum_{j=1}^{Ni}\sum_{m=-6}^{6} q_{6m}(i)q_{6m}^*(j), \tag{2}$$

where the asterisk indicates complex conjugation. Therefore, the $q_6q_6^*$ parameter can be used to accurately distinguish between the amorphous and crystalline phases at the atomic level in semicrystalline polymer systems [66,67]. Going one step further, in this study, the distribution of the $q_6q_6^*$ parameter was used to determine the boundaries of the transient and bulk regions of the polymer crystal. The determination of the spatial area of the crystal using the $q_6q_6^*$ distribution and its correlation with the thermal properties are discussed in Section 3.

2.3. Derivation of thermal conductivity of the microstructure

Non-equilibrium molecular dynamics (NEMD) simulations were performed to estimate the thermal conductivity of the modeled lamellae. The lamellar structure was individually replicated thrice in each direction, and additional equilibration was performed in the NVT ensemble for 5 ns. Subsequently, 1-nm-thick slabs were set at either end along the replicated directions as constrained regions where atomic motion was completely excluded. To establish a temperature gradient in the simulation model, a 1-nm-thick zone was created at the two ends in the longitudinal direction as the heat source and heat sink, respectively, with the thermal energy added and subtracted at a constant rate of 0.02 kcal/mol fs (see Fig. S3 for the energy deviation of the heat sink and source). All zones except for the heat supply and heat sink regions were time-integrated for 20 ns in the NVE ensemble; this duration was deemed sufficient for generating a converged temperature gradient within the microstructure. Under these conditions, the thermal conductivity (K) in the direction of the heat flow could be calculated based on Fourier's law.

$$K = \frac{dE/dt}{A\Delta T/\Delta L}, \tag{3}$$

where A is the cross-sectional area along the longitudinal direction, dE/dt is the rate of energy exchange between the heat source and heat sink, and $\Delta T/\Delta L$ is the temperature gradient. To consider the temperature dependence of the thermal conductivity, equilibrium temperatures of 300, 325, and 350 K were selected.

3. Results and discussion

The procedures for the theoretical modeling to predict the macroscopic thermal conductivity of semicrystalline polymer are as follows. The evolution of the lamellar morphology during isothermal crystallization was analyzed based on structural-order parameters. The chain-parallel and -perpendicular thermal conductivities depending on the order parameter were derived from NEMD simulation. Subsequently, polymer crystals were divided into three phases of transient–bulk–transient, and the thermal-conductivity distribution in each region was obtained. The effective thermal conductivity of the polymer crystal depending on the lamellar thickness and orientation angle was also determined. Finally, the thermal conductivity of the entire semicrystalline system was obtained based on Maxwell's effective medium

theory and Series model. The prediction was quantitatively validated with the reported experimental literatures.

3.1. Morphological analysis of semicrystalline PE microstructure

The evolution of the lamellar structure during isothermal crystallization was monitored (Fig. 2(a)). The PE polymer melted completely at the extremely high temperature used (700 K); thus, the initial model represented a completely amorphous arrangement with $q_6q_6^*$ values close to zero. However, events in which adjacent polymer chains line up next to each other occasionally occurred over time. At this time, the interaction by potential energy between the two segments dominates over thermal energy, which leads to the gradual growth of regularly ordered microstructures in the model. Specifically, the PE crystals that started forming at 50 ns continued to connect infinitely across the *yz*-plane boundary until 100 ns, generating periodic lamellae along the *x*-axis.

The progress of crystallization was clearly identified from changes in the $q_6q_6^*$ distribution of the beads constituting the PE chains (Fig. 2(b)). The results obtained at each simulation time point were interpolated to kernel density estimates. The profile first exhibited a unimodal distribution centered at zero but switched to a bimodal shape with crystal growth, suggesting that amorphous and crystalline phases coexisted distinctly in the PE microstructure. Therefore, the states of the individual phases of the semicrystalline system and their ratios could be rigorously determined by defining an appropriate cutoff criterion based on the $q_6q_6^*$ distribution. Additionally, minor changes were observed in the profile for elapsed times greater than 80 ns, indicating completion of the crystal growth in the given space (the changes in the $q_6q_6^*$ distribution during lamellae construction are given in Fig. S1).

It is worth noting that the thermal-energy-induced vibrations of the constituent atoms significantly affect the $q_6q_6^*$ calculation. Therefore, to focus only on the morphological features of the semicrystalline structure, the PE model that attained the desired crystallinity was further relaxed in the NPT ensemble (300 K, 1 atm) for 5 ns and then rapidly quenched in the NVT ensemble (0.1 K) for 1 ns. The $q_6q_6^*$ distribution of the fully crystallized model was then obtained at a cryogenic temperature (Fig. 2(c)). As the disturbance caused by the thermodynamic energy disappeared, two distinct peaks and an intermediate transient region emerged in the profile. Previous studies have typically used a 95 % confidence level for the Gaussian distribution of the upper peaks to define the crystalline phase of the microstructure [66,68]. However, in the coarse-grained model adopted in the present study, the beads in the transient region ($q_6q_6^* \cong 1.0$) formed a remarkably regular arrangement at the lamellar surface (Fig. 2(a)). Therefore, all the regions except the 99 % confidence level of the Gaussian curve at the lower peak (near zero) were considered the crystalline area, with the $q_6q_6^*$ cutoff value being 0.55.

The spatial distributions of $q_6q_6^*$, local density, and orientation of the polymer chain along the lamellar stacking direction (that is, the *x*-axis) were subsequently analyzed (Fig. 2(d)). Because the crystalline and amorphous structures were clearly separated in the implemented semicrystalline model, $q_6q_6^*$ varied sinusoidally along the *x*-direction. The higher the degree of order of the polymer backbone, the more condensed the microstructure; therefore, the mass density behaved similarly to $q_6q_6^*$. Notably, in the defined crystalline region ($q_6q_6^* >$

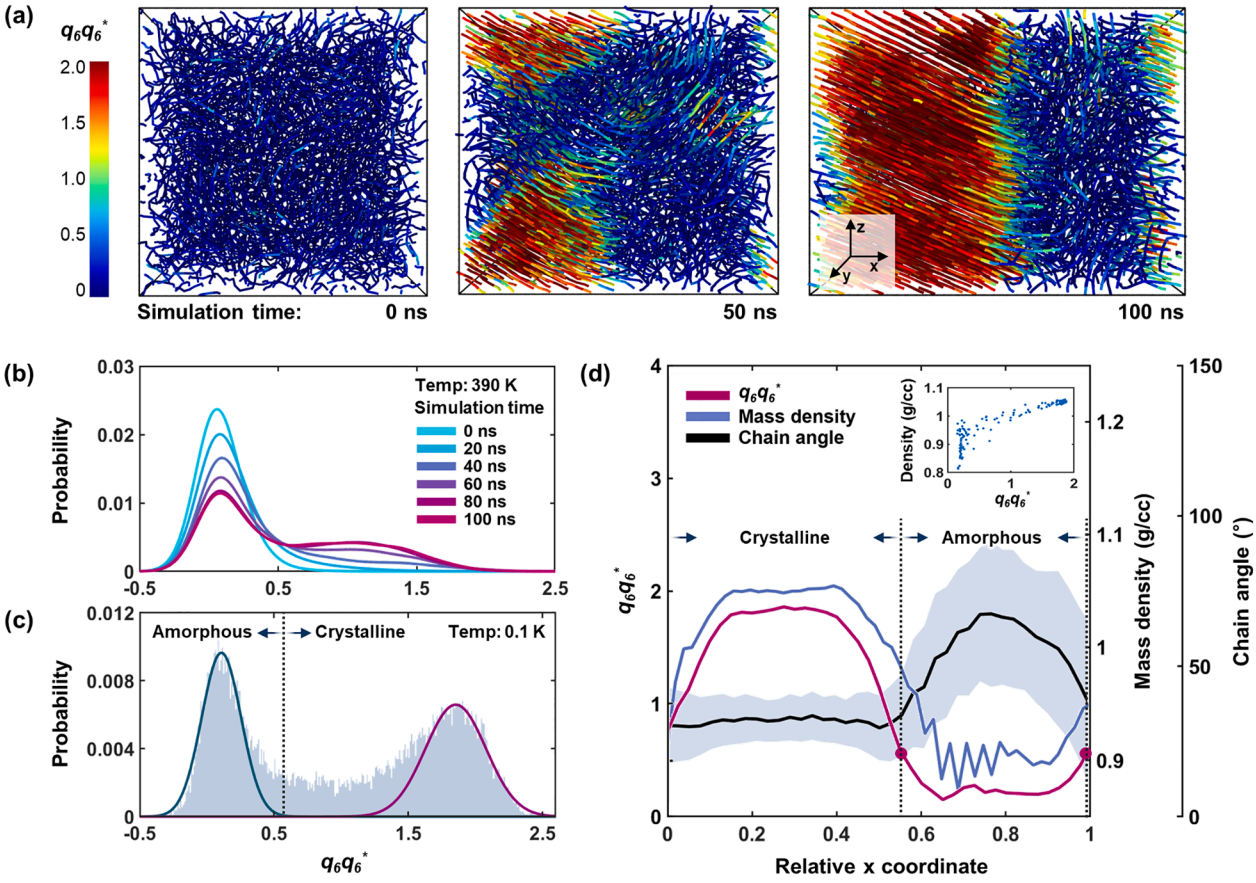


Fig. 2. Morphological analysis of semicrystalline PE during isothermal crystallization. (a) Simulated time evolution of polymer lamellae. The configurations are colored based on their $q_6q_6^*$ order parameters. (b) Distribution of $q_6q_6^*$ order parameter as the simulated crystallization progressed. (c) Distribution of $q_6q_6^*$ after quenching to 0.1 K, with each peak in the profile fitted to a Gaussian distribution. (d) Spatial variations of $q_6q_6^*$, mass density, and backbone-chain orientation angle along the lamellar axis (that is, the *x*-axis). In the $q_6q_6^*$ profile, the two dots represent the datapoints corresponding to a value of 0.55. The shaded area denotes the standard deviation of the chain-angle profile. The correlation between $q_6q_6^*$ and mass density is shown in the inset.

0.55), $q_6q_6^*$ profile can be divided into three distinct stages leading to increases, plateaus, and decreases. In other words, even within crystalline domains, the structural properties varied significantly depending on whether the polymer chain existed within the complete bulk or in the interfacial region with the amorphous material. This highlighted the need for spatially analyzing the thermal conductivity of the crystalline lamellae; these aspects are addressed in the next section.

The polymer chains maintained a constant orientation of ~30° in the crystalline region, which increased with the transition into the amorphous region. The backbone angle varied negligibly throughout the defined crystalline region, establishing the validity of the applied criterion based on $q_6q_6^*$. Other structural features including the densities of the crystalline or amorphous phases and polymer chain orientation also showed good agreement with the reported experimental data of melt-crystallized PE (Table S1 in Supplementary Data) [69–72]. The only difference is the thickness of the crystalline region, but this can be rationalized in that a sufficient convergence of thermal behavior is observed in the model considered. Overall, the established microstructural model was directly used to reproduce the structural properties of crystallized PE.

3.2. Measurement of local thermal conductivity of crystalline lamellae

NEMD simulations were performed to evaluate thermal conductivity as a function of the polymer alignment (Fig. 3(a)). When thermodynamic energy exchange was achieved in the NEMD simulations, no variation was observed in the temperature gradient during the last 4 ns (Fig. S4). In other words, the entire system was in a steady state during this period; therefore, the calculated heat conductivity was also consistent. Furthermore, when the converged temperature gradients of the PE microstructure in the longitudinal direction were analyzed (Fig. 3 (b)), clear differences were observed in the heat flow rates for each phase in the lamellar stacking direction (that is, the x-axis) with alternating crystalline and amorphous regions. Furthermore, the temperature deviation in the longitudinal direction was minimal across the crystalline domain, indicating the robust thermal-transporting ability of the regularly arranged polymer structure. In contrast, the temperature decreased monotonically in the direction perpendicular to the lamellar stack; these datapoints were obtained by averaging the data over the y- and z-axes.

The calculated temperature gradient of the semicrystalline polymer was further examined by monitoring the change in thermal conductivity with respect to the order parameter $q_6q_6^*$. A more detailed description of the local thermal conductivity measurements is provided in Supplementary Note 2 and Figs. S5, S6 of Supplementary Data. The aforementioned analytical strategy clarifies the correlation between thermal conductivity and the degree of alignment of the polymer chains forming the microstructure.

One of the considerations in the NEMD simulations is the size effect caused by limited length. When the model length is insufficient, the resulting thermal properties is generally underestimated. In this regard, Subramanyan et al. and He et al. showed that this size dependence almost disappears when the model length reaches 20 nm [38,73]. The model used in this study was replicated thrice for the NEMD simulations, reaching axial and transverse lengths of 37.2 and 26.1 nm, respectively. In other words, the model length is considered sufficient for accurate thermal conductivity analysis. Additionally, we measured the axial thermal conductivity of the PE lamellae of various lengths at 300 K (Fig. S7), whereby the thermal conductivity results converge significantly at a model length of approximately 25 nm.

The thermal conductivity of the polymer lamellae was investigated as a function of $q_6q_6^*$ (Fig. 3(c)). In the direction parallel to the lamellar stacking, a significant change in thermal properties was observed depending on the backbone chain order. This is consistent with the findings of Sæther et al. [74], who revealed an exponential relationship between the local thermal conductivity estimated along the lamellar orientation and the mass density. Furthermore, the mass density and $q_6q_6^*$ exhibited a highly linear correlation (Fig. 2(d), inset). Based on

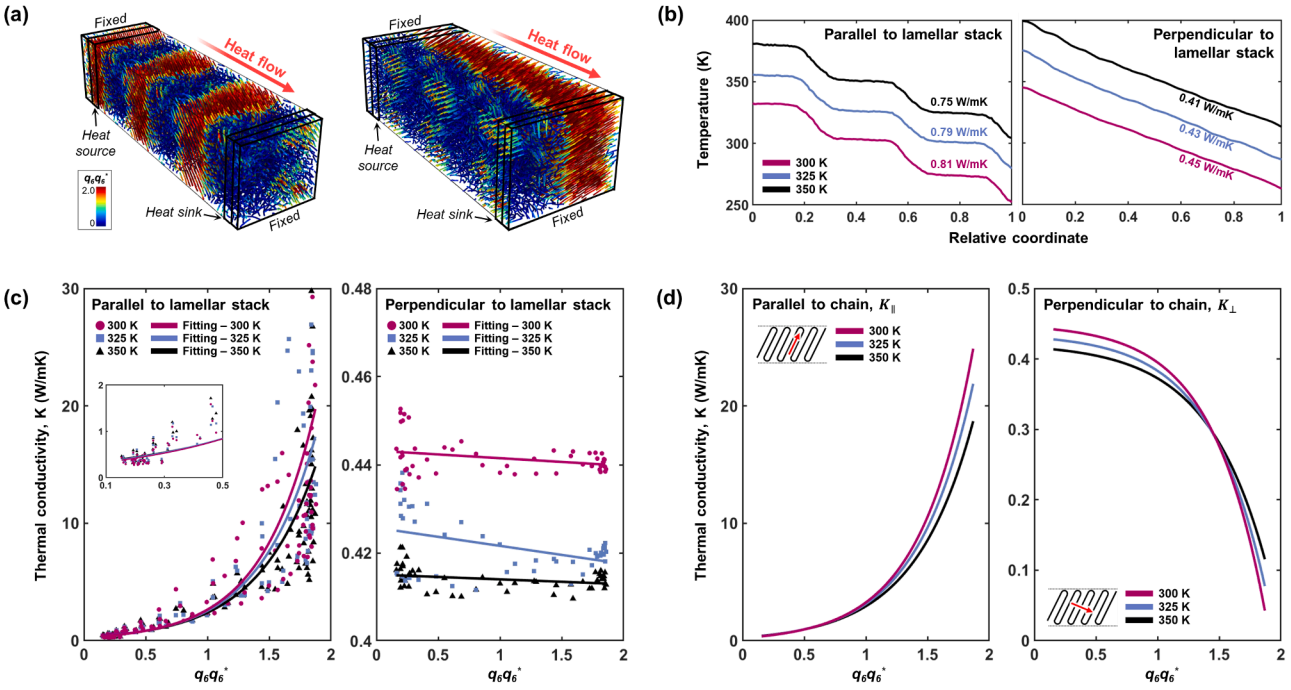


Fig. 3. Thermal conductivity of the modeled PE microstructure. (a) Schematic of the NEMD simulations performed parallel (left) and perpendicular to the lamellar stack (right). The configurations are colored based on their $q_6q_6^*$ order parameters. (b) Steady-state temperature gradient along the longitudinal axis of the models. The thermal conductivity for each scenario is calculated by the linear fitting of the entire temperature-gradient profile. (c) Correlation between the order parameter $q_6q_6^*$ and thermal conductivity. In the results obtained parallel to the stacking direction, the inset shows the region with low $q_6q_6^*$ values (0.1–0.5). (d) Correlation between the order parameter $q_6q_6^*$ and thermal conductivity when the latter is decomposed parallel (left) and perpendicular (right) to the main axis of the polymer crystal.

these findings, the thermal conductivity data obtained at the given temperatures were fit using an exponential function (see Table S2 for the coefficients of the exponential functions).

The fit exponential functions indicated that the thermal conductivity, which was ~0.37 W/m K in the amorphous phase ($q_6q_6^* \cong 0.15$), drastically improved to 20 W/m K in the crystalline phase ($q_6q_6^* \cong 1.82$). The observation that the degree of alignment of the polymer microstructure has a significant influence on the thermal properties of the overall microstructure has also been addressed in previous studies [75]. One new finding is that even within the region determined to be crystalline ($q_6q_6^* > 0.55$), the thermal conductivity of PE is not constant and varies greatly over a wide range. This is evidence that the intrinsic thermal properties of crystalline chains are greatly affected as they are exposed to a surface or interact with a disordered amorphous phase (also refer Fig. S8 for vibration analysis for $q_6q_6^*$). Therefore, the thermal-transporting ability of polymer chains constituting the crystalline phase must be carefully classified and evaluated according to their spatial location within the lamellae.

Another important finding is the gradual decrease in the axial thermal conductivity with temperature. Shrestha et al. noted that the decrease in the thermal conduction of PE nanofibers during heating is ascribed to the Umklapp scattering of phonons in polymer crystals [76]. In other words, random vibrations of atoms magnified by thermal energy cause phonon collisions and disrupt the heat flow. For further validation, we examined the displacement distributions of the well-organized crystalline chains during temperature rise and verified that oscillation anisotropy gradually decreased with the supply of thermal energy (Fig. S9). In other words, the constituent atoms stand out owing to their nondirectional vibration, which explicitly accounts for the Umklapp phonon scattering.

On the other hand, thermal properties of PE perpendicular to the stacking direction were independent of $q_6q_6^*$ and decreased monotonically with increasing applied temperature. Furthermore, the overall magnitude of the conductivity was extremely low (~0.43 ± 0.02 W/m K). This thermal conductivity anisotropy is plausible, given that the thermal transport in polymer systems is mainly achieved through covalently linked backbone chains. Even in regularly ordered polymer crystals, thermal transfer between parallel-aligned chains depends on the potential energy achieved through noncovalent bonds. Therefore, the thermal conductivity between adjacent chains within the crystalline phase was remarkably low, similar to that of the amorphous phase.

Chains in melt-crystallized polymers have a tilt angle of 30°–35° along the stacking direction [71]. This structural feature was satisfactorily reproduced in the crystalline model reported herein. Therefore, the effective thermal conductivity along the principal axes of the crystal within the microstructure can be effectively characterized using the model. As the PE crystal is orthorhombic, the thermal conductivity tensor along the principal axis can be expressed as:

$$K = \begin{bmatrix} K_{\parallel} & 0 & 0 \\ 0 & K_{\perp} & 0 \\ 0 & 0 & K_{\perp} \end{bmatrix}, \tag{4}$$

where $K_{\perp}$ and $K_{\parallel}$are the thermal conductivity along the chain at the perpendicular and parallel directions, respectively. Using Einstein notation, the thermal conductivity in each direction can be described as follows [77]:

$$\begin{aligned} &K(\theta,\varphi) = K_{ij} l_i l_j, \ (i,j = 1,\ 2,\ or\ 3), \\ &l_1 = \sin\varphi\cos\theta, l_2 = \sin\varphi\sin\theta, l_3 = \cos\varphi, \end{aligned} \tag{5}$$

where l_1, l_2, and l_3 are the direction cosines; and θ and φ represent the azimuthaangle and polar angle in spherical coordinates, respectively. Therefore, the angle-dependent thermal conductivity can be obtained as

$$K(\theta,\varphi) = K_{\parallel}\sin^2\varphi\cos^2\theta + K_{\perp}\sin^2\varphi\sin^2\theta + K_{\perp}\cos^2\varphi\cdot \tag{6}$$

When perpendicular to the stacking direction, all $q_6q_6^*$ values were assumed to correspond to constant conductivities and decrease at regular intervals depending on the applied temperature. The calculated thermal conductivities at 300, 325, and 350 K were 0.4415, 0.4278, and 0.4141 W/m K, respectively.

An analysis of the $q_6q_6^*$-dependent thermal conductivity components along the principal axes of the crystal (Fig. 3(d)) indicated that the heat conduction behavior along the direction parallel to the chains was almost identical to that of the stacking direction. One distinguishing feature was the substantial change in the thermal conductivity according to $q_6q_6^*$. Additionally, the peak value of the profile quantitatively appeared in the experimentally observed conductive region of highly drawn PE microfibers (~20 W/m K [22]). In contrast, the thermal conductivity of the chain along the vertical direction decreased exponentially with increasing $q_6q_6^*$, suggesting that structurally unstable polymer chains in the transient region tended to scatter more thermals in the radial direction than the chains in the bulk. Finally, the difference associated with the direction of the thermal conductivity component decreased with increasing temperature. This is a self-evident result as thermal energy induces a greater nondirectional vibration of the constituent atoms.

Importantly, the aforementioned findings suggest that the local thermal properties of the polymer chain vary significantly depending on its position within the crystal domain. To rigorously predict the nanoscale thermal performance of semicrystalline systems, comprehensive modeling of the two states of the crystal—the transient and bulk regions—is critical. In this respect, the $q_6q_6^*$ parameter and the structural properties of the polymer chain considerably influence the effective thermal conductivity of the entire polymer crystal system. In the next section, an analytical model for quantitatively evaluating the thermal conductivity of semicrystalline systems and the associated design strategy are described with respect to aspects associated with the lamellar morphology, including thickness, orientation, and crystallinity.

3.3. Characterization of effective thermal conductivity of polymer lamellae

An analytical model of crystal lamellae was devised based on the rule of mixtures. From a geometrical perspective, crystal lamellae can be considered as nanoscale laminate structures with a transient–bulk–transient configuration (Fig. 4(a)). The covalent bonds of the polymer chains across the transient and bulk domains are continuous. Therefore, the rapid temperature change occurring at the discontinuous interface of dissimilar materials can be neglected. This temperature continuity occurs presumably owing to the identical chemical compositions of the transient and bulk regions, leading to marginal interfacial effects. Similarly, the interface between the transient and bulk domains can be considered to be fully intact. These properties permit the use of a simple mixing rule to estimate the effective thermal conductivity of the crystal lamellae.

The spatial distributions of $q_6q_6^*$ and the thermal conductivity tensor for L-nm-thick lamellae were examined (Fig. 4(b)). The obtained $q_6q_6^*$ profile could be divided into nearly parallel and tilted regions, which were considered the bulk and transient domains, respectively. Specifically, in the lamellar model, the $q_6q_6^*$ profile converged after a distance of ~2 nm from the amorphous domain (Fig. 2(d)). Therefore, the transition region was set at a constant thickness of 2 nm at the end faces of the crystal; notably, this value is comparable to that estimated by Rungswang et al. via small-angle X-ray scattering (0.8–1.4 nm) [72]. Furthermore, we also confirmed that a 2 nm thickness of the transient region could be extended to typical semicrystalline polymer systems (Supplementary Note 3 and Fig. S10). Additionally, the $q_6q_6^*$ value for the bulk domain was constant (1.82); however, in the transient region, it was obtained by fitting the profile of the devised lamellar model with a quadratic function (see Fig. S11 for the calculation procedure).

Using the correlation between $q_6q_6^*$ and the local conductivity defined in the previous section, the spatial distribution of the thermal

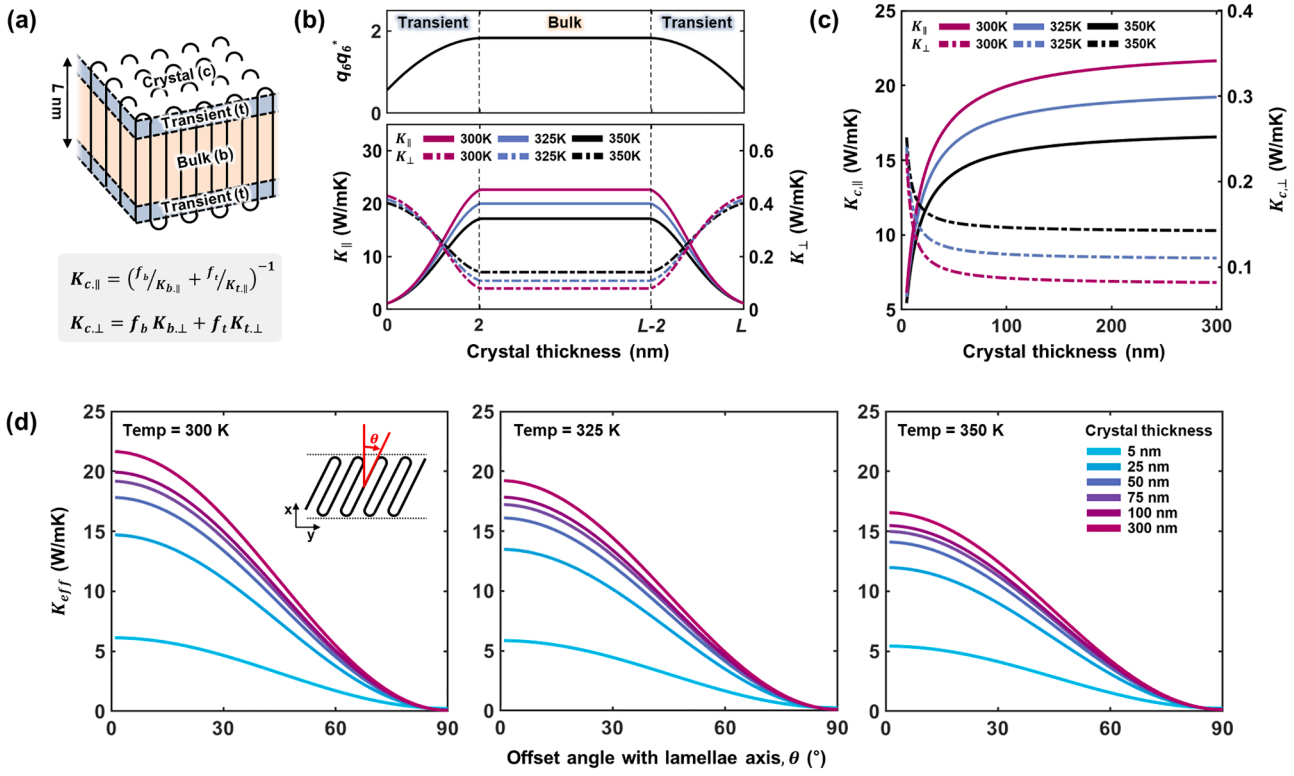


Fig. 4. Mathematical modeling of the thermal conductivity of crystal lamellae. (a) Schematic of the lamellar structure with a transient layer on its surface. (b) Spatial distributions of the order parameter $q_6q_6^*$ and thermal conductivity for an L-nm-thick lamella. (c) Thermal conductivity of polymer crystal based on the lamellar thickness. (d) Effective thermal conductivity of lamellae as functions of the crystal thickness and chain orientation.

conductivity within the crystal lamella was quantitatively estimated. The lamellar thickness of melt-crystalline PE (~12–21 nm) [72] can be extended up to 800 nm through fibrillation by mechanical processing [78]. Therefore, the effective thermal conductivity tensors of the crystal were evaluated with respect to the lamellar thickness (Fig. 4(c)). The Reuss and Voigt models were adopted to predict $K_\parallel$ and $K_\perp$, respectively. The results suggested that increasing the crystalline thickness was key to maximizing the chain-directed thermal transport and anisotropy in thermal conductivity. The thicker the crystal, the larger the volume fraction of the bulk crystal phase, which weakened the surface effect. Accordingly, the thermal conductivity tensor of the lamella showed significant convergence at a thickness of ~100 nm. Similar tendencies were observed at all the selected temperatures.

Variations in the effective thermal conductivity of the lamellae owing to offset alignment were analyzed thereafter (Fig. 4(d)). The crystal chains were assumed to rotate in the xy-plane and thus remain perpendicular to the z-axis. The thermal conductivity tensor of the lamellae could be evaluated using Eq. (6). Consequently, the thermal properties were found to deteriorate monotonically with increasing offset angle, consistent with previously reported experimental observations [79]. Additionally, the applied temperature and crystal thickness significantly affected the thermal conductivity of the lamellae; the lower the applied temperature and the thicker the crystal, the higher the effective thermal conductivity of the lamellae.

3.4. A model describing the effective thermal conductivity of semicrystalline polymer systems

This section presents a scheme for quantitatively predicting the effective thermal conductivity of a semicrystalline system based on the crystal morphology. Thermal transport was comprehensively analyzed according to the subjected temperature, crystallinity, lamellar thickness, and crystal orientation. Two cases were considered in terms of the crystal growth: isotropic and preferential. Based on the morphological observations, a theoretical model suitable for each case was devised, and the results were verified using experimental data.

Modeling starts with an understanding of a spherulite structure, a typical crystal configuration of melt-crystallized polymers. Essentially, microstructures of the alternating crystalline and amorphous phases grow radially in all directions to form spherical aggregates (Fig. 5(a)). Moreover, the thermal conductivity of a single lamella is highly anisotropic. However, as the lamellae grow uniformly along the radius of the sphere, the resulting structure becomes isotropically homogenized. As the growth progresses, adjacent spherulites overlap, and the interface entirely comprises long polymer chains.

To apply the aforementioned established concept to the MD simulation data acquired in the present study, a model proposed by Choy and Young based on Maxwell's effective medium theory was introduced [80]. For crystallites that do not have a preferred orientation, the crystallinity-dependent effective thermal conductivity is calculated as follows:

$$\frac{K_e - K_a}{K_e + 2K_a} = X\left[\frac{2}{3}\frac{K_{c\perp}/K_a - 1}{K_{c\perp}/K_a + 2} + \frac{1}{3}\frac{K_{c\parallel}/K_a - 1}{K_{c\parallel}/K_a + 2}\right], \tag{7}$$

where K_e, K_a, $K_{c\perp}$, and $K_{c\parallel}$ indicate the thermal conductivities of the entire system, amorphous phase, crystalline phase perpendicular to the chain direction, and crystalline phase along the chain direction, respectively; and X represents the crystallinity of the semicrystalline system.

As stated in the previous section, the thermal resistance between the amorphous and crystalline domains was negligible, and the temperature gradient was continuous (Fig. 3(b)). Therefore, the effective thermal conductivity of the microstructure was determined based on the mixing rules for the amorphous and crystalline fractions. One consideration in applying the model is the thermal conductivity of the amorphous region. At room temperature, the thermal conductivity of PE in the fully

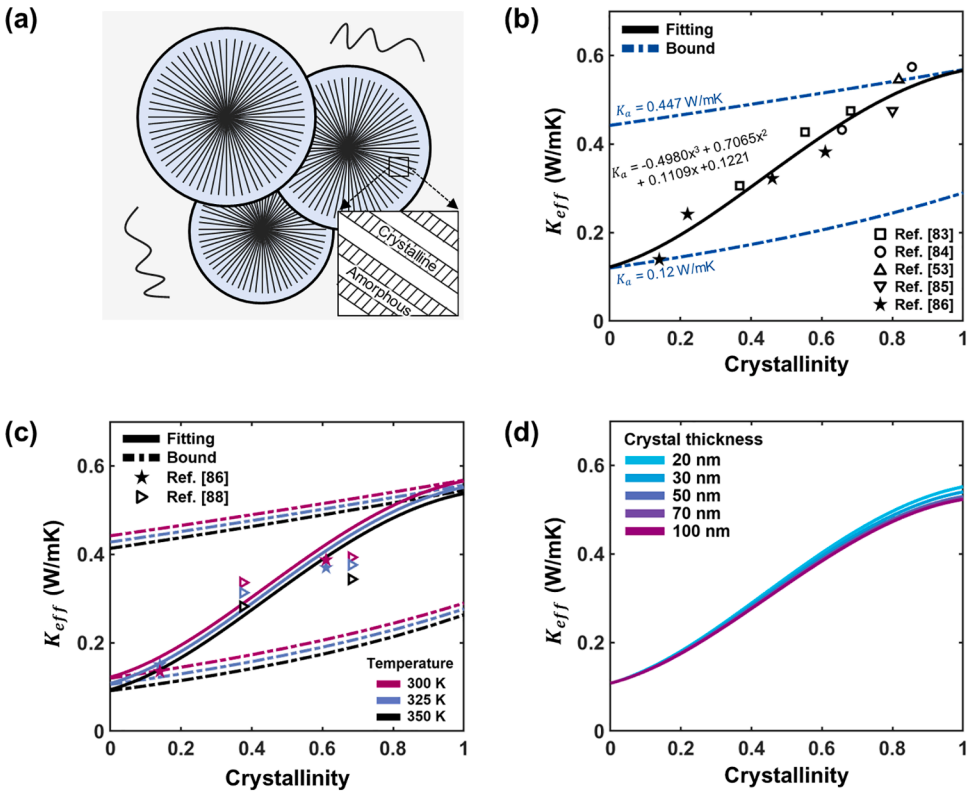


Fig. 5. Estimation of the effective thermal conductivity of melt-crystallized PE microstructures. (a) Schematic of polymer spherulites. (b) Crystallinity-dependent effective thermal conductivity at room temperature. (c) Thermal conductivity of isotropically crystallized polymer for different temperatures. (d) Influence of crystal lamellar thickness on thermal conductivity.

disordered state is 0.12 W/m K [81], whereas that of amorphous chains in the inter-lamellar region reaches 0.447 W/m K (Fig. 3(d)). Considering the gradual incorporation of fully disordered chains into those of interlamellar with increasing crystallinity, each value can serve as a lower and upper bound for the thermal conductivity.

The crystallinity-dependent effective thermal conductivities were compared with those obtained experimentally (Fig. 5(b)) [53,82–86]. Considering that a typical melt-crystallized lamella is 20 ± 2 nm thick [87], the corresponding $K_{c\|}$ and $K_{c\perp}$ values (13.52 and 0.1176 W/m K) were used for the calculations. The experimentally obtained thermal conductivity data were within the range of values predicted by the model. As expected, previously reported data associated with low crystallinities were close to the lower limit, whereas the opposite was true for high crystallinities. Additionally, by fitting and inserting K_a obtained from the reference data, a crystallinity-dependent thermal conductivity curve was constructed using the model.

Changes in the thermal conductivity with temperature were analyzed for the spherical crystallization system (Fig. 5(c)). $K_{c\|}$ and $K_{c\perp}$ were adjusted for each temperature, whereas K_a was assumed to linearly decrease by 0.014 W/m K for every 25 K increase in temperature, based on the results shown in Fig. 3(d). As the temperature increased and thermal scattering intensified, the thermal conductivity of the semi-crystalline polymer decreased gradually. Moreover, the literature-derived data for each temperature fit well within the proposed bounds [86,88], and each profile successfully predicted the temperature-dependent thermal conductivity at a given level of crystallinity.

The lamellar thickness of isotropically crystallized polymers can be increased to 45 nm by manipulating the cooling rate, thermal annealing temperature, and pressure during processing [89,90]. Motivated by this finding, the influence of the lamellar thickness on the effective thermal conductivity was evaluated (Fig. 5(d)). Thicknesses of 20–100 nm was considered, and the environment was assumed to be ambient (300 K). Interestingly, increasing the lamellar thickness had a minor effect on the thermal conductivity of the overall system. Even when the crystal thickness was increased five-fold (from 20 to 100 nm), the effective thermal conductivity showed a maximum difference of only 5 %. In other words, the strategy of improving the thermal properties of spherically crystallized systems by increasing $K_{c\|}$ appears to be unfavorable. This result also agrees with Choy and Young's finding that $K_{c\|}$ varies minimally prior to exceeding 10 W/m K [80]. Phenomenologically, these results suggest that the thermal transport efficiency provided by the crystalline phase rapidly saturates with increasing thickness. In other words, the thermal properties of the entire system depend only on $K_{c\perp}$, which is inversely proportional to the thickness.

The aforementioned results highlight the factors that should be addressed to substantially improve the thermal properties of nondirectionally crystallized polymers. Strategies employed to increase the percentage of crystallinity within amorphous-phase domains or to increase the degree of alignment of polymer segments (that is, improving K_a) are effective in enhancing the thermal conductivity. Another promising approach involves strengthening the interchain interactions, such as partial dipole formation in poly(vinylidene fluoride) [91]. The strong binding force permits efficient thermal transport along the vertical direction of the chain, thereby helping significantly in improving the thermal conductivity of the overall system. This leads to an improvement in the overall thermal conductivity in the form of increasing $K_{c\perp}$. In contrast, further alignment and enrichment of the crystal phase (that is, improving $K_{c\|}$) is somewhat ineffective at improving the overall thermal conductivity.

Mechanical processes such as extrusion and stretching allow polymers to have specific orientations in both the crystalline and amorphous states (Fig. 6(a)). Accordingly, phonons traveling in the stretching direction (that is, dominant in thermal vibrations) encounter alternating crystalline and amorphous phases. Considering the layers of individual phases aligned perpendicular to the heat flow, the thermal conductivity

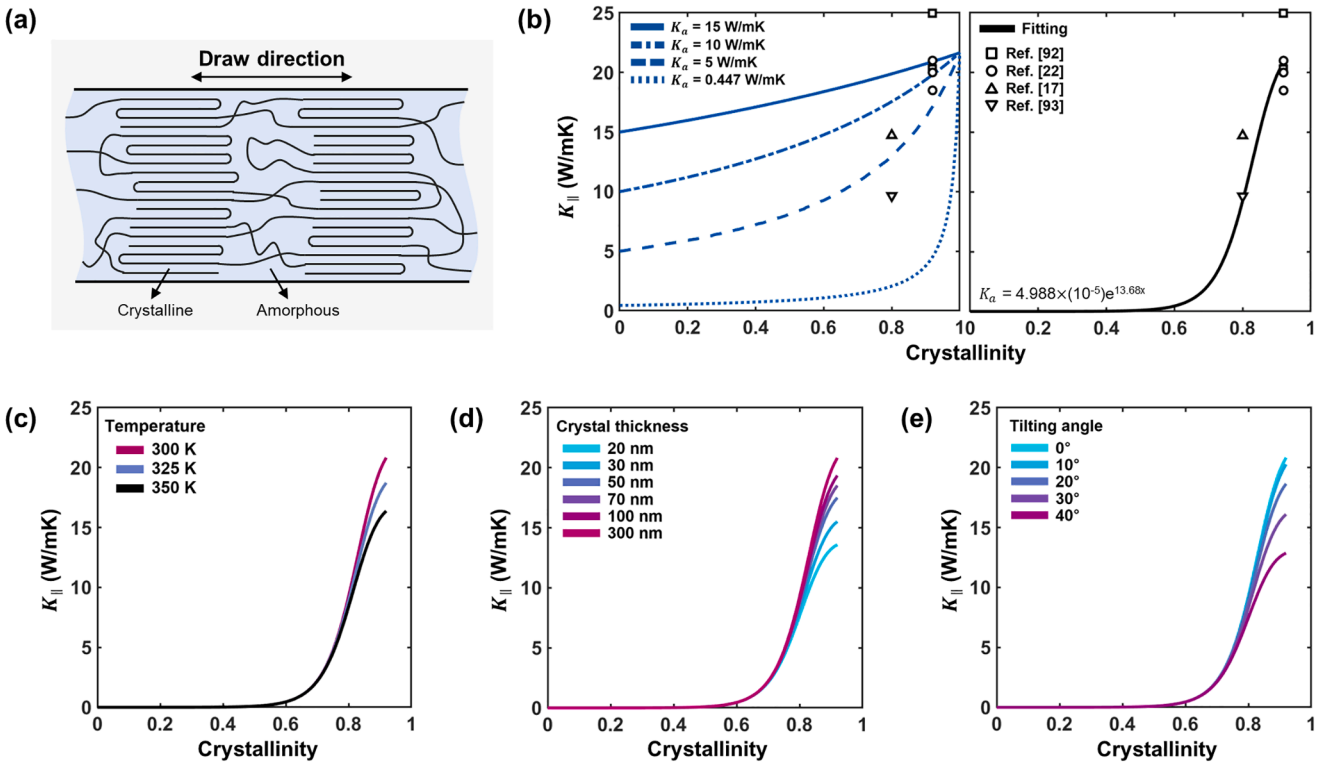


Fig. 6. Prediction of the effective thermal conductivity of crystallized PE with specific orientations. (a) Schematic of a highly drawn polymer microstructure. (b) Thermal conductivity as a function of crystallinity (left), calculated for fitted K_a values (right). Effects of (c) temperature, (d) crystal thickness, and (e) lamellar orientation on the thermal conductivity of crystalline polymers.

of the system was predicted using a simple series model. The crystallinity-dependent thermal conductivity along the axial direction can be obtained as follows:

$$K_{\parallel} = \left(\frac{X}{K_{c\parallel}} + \frac{1-X}{K_a}\right)^{-1}. \tag{8}$$

Similar to the case of spherulites, a key aspect of modeling the oriented crystals was defining the thermal conductivity of the amorphous region. According to Xu et al., the effective thermal conductivity of the amorphous phase in a drawn polymer increases exponentially with the draw ratio and can reach 15 W/m K [16]. Therefore, the crystallinity-dependent thermal conductivity profiles were obtained by considering various K_a values (Fig. 6(b), left). The temperature was assumed to be 300 K, and the thermal conductivity of the crystal domains was obtained using a sufficiently large thickness (300 nm), considering the fibrillation of polymer chains during processing. Additionally, the molecular chains were assumed to be oriented parallel to the stretching direction.

Notably, the change in thermal conductivity with crystallinity was dramatic (Fig. 6(b)). At a low crystallinity of ~0.8, previously reported experimental data on the thermal conductivity of PE microfibers [17,22,92,93] were close to the profile of K_a = 5 W/m K. However, with increasing crystallinity, the conductivity approached the profiles corresponding to K_a = 10–15 W/m K. These findings suggest that the K_a value was not constant but varied continuously as crystallization progressed. Based on these observations, K_a was expressed as a function of crystallinity in our model using the data presented by Xu et al. [16] (Fig. S12). The thermal conductivity profile fit using the aforementioned process (Fig. 6(b), right) indicated that the series model satisfactorily reflected the nonlinearity of the effective thermal conductivity of highly oriented polymers at different crystallinity levels.

Using the considered model, the thermal conductivity of the system was predicted according to crystal morphological features: the exposure temperature, thickness, and lamellar tilt (Fig. 6(c)–(e)). To focus solely on the influence of the crystalline structure, the thermal conductivity of the amorphous phase was assumed to be constant under all conditions. The analysis results suggest the following. First, the higher the experienced temperature, the greater the decrease in the axial conductivity of the system, thereby promoting thermal scattering. Second, unlike in spherulites, the thermal properties of highly oriented polymers are largely determined by the lamellar thickness. This is broadly consistent with previous findings on the fibrillation of polymer chains leading to dramatic changes in thermal conductivity [94]. Third, even small changes in orientation significantly affect the thermal conductivity of the entire system. Laramée et al. showed that the overall orientation of polymer fibers could be adjusted from 20° to 35° depending on the solvent used during electrospinning [95]. According to our analysis, the aforementioned state change will likely alter the effective thermal conductivity by ~30 % (predicted values: 18.66 and 14.49 W/m K at 20° and 35°, respectively).

An interesting finding of the present study is that $K_{c\parallel}$, which is influenced by the crystal morphology, significantly affects the thermal conductivity of crystallized PE. Indeed, meticulously aligning polymer chains to achieve high levels of thermal conductivity is a well-established polymer design strategy. Nevertheless, the present study is important because it theoretically demonstrates the effects of the crystallinity and anisotropy of polymer microstructures on thermal conductivity. In particular, it validates a new approach to devising a thermal conductivity design modeling that can be directly matched with experimental observations.

The inherent properties of polymers, including high processability, electrical insulation, low cost, and chemical resistance, promote their application in thermal management systems. Typical industrial applications include packaging for LEDs, semiconductor nanostructure, battery packs, and thermal protection films for solar panels and displays. Recently, thermally conductive polymers have become promising

candidates for applications that require both high thermal conductivity and material deformability, including wearable electronics and artificial skin. In this regard, this study explains the role of the microstructure of a semicrystalline system in determining its macroscopic thermal conductivity. In particular, the influence of a wide range of physicochemical parameters, including crystallinity, lamellar thickness, and molecular orientation, has been comprehensively explained, thereby providing a theoretical framework for the molecular-level design of thermal conductivity manipulation. Therefore, our research complements the ongoing efforts to develop advanced thermal management systems.

4. Conclusions

In this study, theoretical modeling was performed to estimate the thermal conductivity of semicrystalline polymers depending on their crystal morphology at the molecular scale. PE microstructures with coexisting crystalline and amorphous phases were implemented in a coarse-grained MD environment using isothermal crystallization simulations. Under a constant heat flow, the local thermal response of the crystalline region was evaluated according to the structural order. The difference in the structural stability resulted in the stronger lateral heat scattering in the transient region than in the bulk crystalline region, resulting in the lower axial thermal conductivity. Therefore, crystal lamellar modeling was regarded as a multiphase problem, considering the fraction and orientation of the transitional and bulk regions.

The thermal-conductivity distribution in each phase was specified using the $q_6q_6^*$ molecular order, and the effective properties of the lamellae were evaluated using a mixing model based on geometric observations. In particular, directional thermal conductivity and anisotropy were characterized with respect to the crystal thickness, exposure temperature, and orientation. Lamellar modeling was integrated with homogenization theory and extended to predict the thermal properties of two hierarchical growth structures: spherulites and oriented crystals. The derived estimates from Maxwell and series models demonstrated quantitative agreement with the experimental literatures. In other words, the thermal conductivity of the entire system can be designed bottom-up based on the crystalline morphology of the microstructure.

To our knowledge, this study is the first attempt to quantitatively predict the influence of the morphological properties of the polymer microstructure on heat flow. We successfully modeled the process by which the macroscopic heat conduction characteristics were determined by analyzing the effects of the morphological parameters determined at the molecular scale. Therefore, this report can serve as a theoretical guidepost for the rational selection and design of semicrystalline materials for engineering applications, such as advanced plastic heat sinks and thermal management systems for flexible electronics.

CRediT authorship contribution statement

Hongdeok Kim: Writing – original draft, Visualization, Software, Methodology. **Joonmyung Choi:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijmecsci.2024.109622.

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